COMP3670 / COMP6670: Introduction to Machine Learning
2026 Semester 2 Assignment 1

Instructions

Please read all instructions before you begin attempting the questions. Failure to follow the instructions may result in mark penalty.

- **Deadline:** 21 August 2026, 11:59pm (Canberra time)
- **Submission:**
  - Submit your solutions to all the questions in **one single pdf file** through Gradescope.
  - Start each new part on a new page.
  - You are encouraged to typeset your solutions using LaTeX. If you opt for handwriting, make sure that it is clear and easy to read. Unreadable writing may not be graded.
- **Grading:**
  - There are two questions. The first one has two parts and the second one has three parts. Total marks are 30.
  - After uploading your pdf file in Gradescope, you will be instructed to **assign page numbers to each of the five parts**. **A 10% penalty will be applied for each part you fail to assign page numbers to.** A simple guide for doing so is available here.
  - Justify your answers and show all your steps clearly. Correct numerical answers without supporting justification/steps will receive zero mark.
- **Academic Integrity:**
  - This is an **individual assignment**. You must work on your own.
  - The use of any generative AI tool is **not** allowed.
  - For more detailed guidelines on academic integrity, see Week 1 lecture slides and ANU webpage on academic integrity.

Question 1: Magpies, science and the supervised learning framework (15 marks)

Every spring the ANU *Landscape & Conservation* team receives reports of cyclists being swooped by magpies while riding through campus. The team must decide, for each of the many locations on campus, whether to install a warning sign and whether that sign must be lit at night. Instead of guessing, they decide to attack the problem *scientifically*, and eventually with machine learning. In this question you will take them through both routes of scientific reasoning seen in Lecture 1, and then place the problem inside the supervised learning framework.

Throughout, an **encounter** means one cyclist riding past one magpie territory on one day, and it either results in a swoop or it does not.

(a) Ornithologists, i.e., scientists who study birds, have a well-established *theory*: magpies swoop only during the breeding season in order to defend a nest that contains chicks against animals they perceive as intruders.

(i) State one *hypothesis* about swooping on campus that is implied by this theory and that could be confronted with a record of encounters. Your hypothesis must be a general claim about encounters, not a claim about one particular bird. **[2 marks]**

(ii) Describe *one single* observation that would falsify the hypothesis you stated in (i). **[2 marks]**

(iii) The team records one full spring of encounters and no such observation occurs. A member of the team concludes: *"the hypothesis is now proved"*. Are they correct? **[2 marks]**

(b) The theory of part **(a)** says *why* magpies swoop, but it is far too coarse to say *which* cyclists, on *which* days, at *which* spots will actually be swooped. The team therefore abandons it and takes the inductive approach to science: start from observations, with no theory assumed beforehand, and look for a pattern.

The team decides to record, for each encounter, whether the cyclist passed within 50 metres of a nest and the hour of the day that the encounter occurred. Then, based on this data, they plan to come up with a pattern about how swoops occur, and apply it to determine high-risk areas on campus for sign placement and whether the signs need to be lit.

(i) Express the problem in the framework of learning from data by specifying the input space $\mathcal{X}$, the output space $\mathcal{Y}$, the loss function ℓ you would use and why, the meaning of the target function $f^\star$ in this specific problem, and the form of the dataset $\mathcal{D}_N$. **[2 marks]**

(ii) Propose to the team a way of collecting a sample that is independent and identically distributed, drawn from the population of all the encounters that occur throughout campus during the spring. **[3 marks]**

(iii) Five springs pass. The team has recorded 20,000 encounters, and in every single one of them the outcome agrees with the pattern they extracted: not one counterexample has ever been observed. On this basis the team writes in its report:

> *"The evidence is overwhelming. We have established that the pattern holds for certain: it is a law of magpie behaviour on the ANU campus."*

Are they entitled to write this? Justify your answer, and then state as precisely as you can what the team *is* entitled to claim instead. Your answer should make clear which quantities the weaker claim would have to mention in order to be a meaningful probabilistic statement about future encounters. **[4 marks]**

Question 2: The role of the noise in regression (15 marks)

A bus operator wants to predict the *delay* of a bus on a given run of a fixed route. Write Y for the delay in minutes, and record two features of the run:

$X_1 =$ rainfall during the run (in cm)
$X_2 =$ fraction of the seats that are occupied when the bus is half-way through the route.

An *oracle* knows that the *population* of runs is described by the following statistical model:

$$\boxed{Y \;=\; 3 \;+\; X_1 \;+\; 2X_2 \;+\; \varepsilon}$$

where X_1, X_2 and ε are **independent**, X_1 and X_2 are both Uniform$[0,1]$, and the noise ε represents the randomness of the delay that is not captured by the rainfall and the bus occupancy, since the delay may depend on other factors that are not observed. The probability density function of the noise is

$$f_\varepsilon(e) \;=\; \begin{cases} 1+e, & -1 \le e \le 0, \\ 1-e, & 0 \le e \le +1, \\ 0, & \text{otherwise.} \end{cases}$$

Throughout this question, the loss function is the **squared loss**, $\ell((\mathbf{x}, y), h) = (y - h(\mathbf{x}))^2$.

Facts you may use without proof.

(F1) If U, V and Z are random variables and a, b, c, d are constants, then

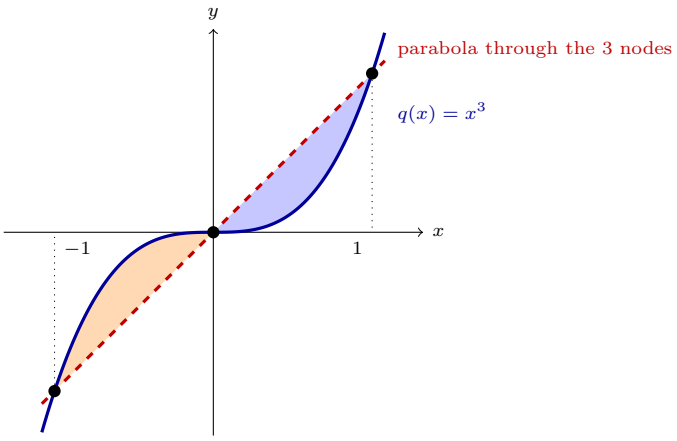
$$\mathbb{E}[a + bU + cV + dZ] = a + b\,\mathbb{E}[U] + c\,\mathbb{E}[V] + d\,\mathbb{E}[Z].$$

Furthermore, if U, V and Z are independent, then

$$\mathsf{Var}(a + bU + cV + dZ) = b^2\mathsf{Var}(U) + c^2\mathsf{Var}(V) + d^2\mathsf{Var}(Z).$$

(F2) Simpson's rule applied to an interval $[a, b]$ returns the *exact* value of $\int_a^b q(x)\,dx$ whenever q agrees on the whole of $[a, b]$ with a polynomial of degree at most 3.

This may look surprising, since Simpson's rule replaces the integrand by a *parabola*, and a parabola is not a cubic. The picture below shows what actually happens, for $q(x) = x^3$ on $[-1, 1]$ with a single parabola ($n = 2$ subintervals, so $\Delta x = 1$ and the nodes are -1, 0 and 1).



The parabola through $(-1, -1)$, $(0, 0)$ and $(1, 1)$ is the straight line $y = x$ (dashed), which is nowhere near x^3. Yet the rule is *exact*: on $[0, 1]$ the line lies *above* the cubic, so Simpson over-estimates the area by the blue shaded amount, and on $[-1, 0]$ it lies *below* the cubic, so it under-estimates by exactly the same amount. The two errors cancel, and the rule returns

$$\frac{\Delta x}{3}\Big[q(-1) + 4\,q(0) + q(1)\Big] \;=\; \frac{1}{3}\big[-1 + 0 + 1\big] \;=\; 0 \;=\; \int_{-1}^{1} x^3\,dx.$$

So Simpson's rule is exact on cubics *not* because the parabola reproduces the integrand, but because the errors it makes cancel out. For degree 2 polynomials the rule is exact because the parabola reproduces the integrand.

Every integral in this question can be evaluated either as the area of a polygon (triangle, rectangle, trapezoid), or by Simpson's rule. You are not expected to use any other integration technique.

(a) Properties of the noise.

(i) Sketch the graph of f_ε and justify why it is a valid density. **[2 marks]**

(ii) Sketch the graph of the function $g(e) = e\, f_\varepsilon(e)$ and explain why $\mathbb{E}[\varepsilon] = 0$ without computing any integral. **[2 marks]**

(b) A student wants to compute $\mathsf{Var}(\varepsilon)$. For this, they write

$$\int_{-1}^{1} e^2 f_\varepsilon(e)\, de = \int_{-1}^{0} e^2 f_\varepsilon(e)\, de + \int_{0}^{+1} e^2 f_\varepsilon(e)\, de,$$

apply Simpson's rule on $[-1, 0]$ and on $[0, +1]$ with $n = 2$ subintervals to approximate the last two integrals, respectively, and then claim that they now know the exact value of $\mathsf{Var}(\varepsilon)$. Is the student's claim correct? Following their reasoning, and correcting it if it is wrong, compute the value of $\mathsf{Var}(\varepsilon)$. *Hint: You might have to apply (F2) at some point.* **[5 marks]**

(c) The operator does not know the model above, and decides in advance to look for a rule among those that combine the two features linearly, that is, to use the hypothesis set

$$\mathcal{H} \;=\; \big\{\, h(x_1, x_2) = a + b\, x_1 + c\, x_2 \;:\; a, b, c \in \mathbb{R} \,\big\}.$$

In this part you will find, by hand, the best rule in $\mathcal{H}$ and will compute its out-of-sample error.

(i) Let $h(x_1, x_2) = a + b\, x_1 + c\, x_2$ be any hypothesis in $\mathcal{H}$. Show that

$$Y - h(X_1, X_2) \;=\; A \;+\; B\, X_1 \;+\; C\, X_2 \;+\; \varepsilon,$$

and identify the constants A, B and C in terms of a, b and c. **[1 mark]**

(ii) Write $W = Y - h(X_1, X_2)$ for the quantity you obtained in (i), so that, by the definition of the out-of-sample error under the squared loss, $E_{\text{out}}(h) = \mathbb{E}[W^2]$. Using $\mathsf{Var}(W) = \mathbb{E}[W^2] - (\mathbb{E}[W])^2$ together with (F1), show that

$$E_{\text{out}}(h) \;=\; \mathsf{Var}(\varepsilon) \;+\; \frac{B^2}{12} \;+\; \frac{C^2}{12} \;+\; \left(A + \frac{B}{2} + \frac{C}{2}\right)^2.$$

Hint: you will need to compute $\mathbb{E}[X_1]$ and $\mathsf{Var}(X_1)$, for which you might apply that $\mathsf{Var}(X_1) = \mathbb{E}[X_1^2] - (\mathbb{E}[X_1])^2$ and use (F2). State clearly where you use the independence of X_1, X_2 and ε. **[3 marks]**

(iii) The three quantities added to $\mathsf{Var}(\varepsilon)$ in (ii) are all squares, hence never negative. Use this to determine the values of a, b and c that make $E_{\text{out}}(h)$ as small as possible, and write down the resulting rule $h^\star$ and the value of $E_{\text{out}}(h^\star)$. How is this value related to the noise, and what does it tell the operator about the best linear rainfall-and-occupancy-based rule they could ever hope to find? **[2 marks]**

Remark: it can be shown that no rule whatsoever, linear or not, achieves an out-of-sample error smaller than the one you obtained in (iii). *The rule $h^\star$ is therefore the target function $f^\star$ of Lecture 2 for this problem, and the hypothesis set $\mathcal{H}$ was a fortunate choice: it contains the best rule there is and there is no approximation error.*